

1. Problem Understanding

- **Ghost nets (derelict gear):** Abandoned, lost or discarded fishing nets and gear (“ghost gear”) continue to fish and entangle marine life long after being lost [9†L12-L17] . They are made of durable synthetic fibers (nylon, PE) and dominate India’s fisheries. Even small, thin nets can trap and kill turtles, fish, and coral for decades [9†L12-L17] [63†L84-L92] .
- **Why hard to detect:** Ghost nets lie on or just above the seafloor and often resemble natural features. They disperse widely, may be partially buried, and cast only subtle acoustic return compared to solid objects [3†L23-L27] . In practice, nets often appear as faint clusters of highlights and shadows on sonar, very hard for human eyes to pick out among rocks and ripples [3†L23-L27] [53†L77-L84] . Locally they blend into seabed patterns. Underwater currents and motion can distort or misalign nets, making their signature irregular. NOAA explains “derelict fishing gear... continues to trap animals, entangle and potentially kill marine life” [9†L12-L17] – a big problem but *nearly invisible* until it’s too late.
- **How side-scan sonar (SSS) works:** SSS tows a transducer (“towfish”) that emits fan-shaped sound pings sideways, mapping a strip of seabed on each side of the vessel. The acoustic backscatter is rendered as grayscale imagery: objects produce bright highlights (strong return) and dark acoustic shadows behind them [53†L77-L84] . Along-track mosaics reveal seafloor texture and objects (see Fig.2 in [13]). In short, side-scan is an **acoustic imaging system** that creates 2D “pictures” of the bottom [6†L293-L302] . Because sound penetrates turbid water, SSS can survey in murky, deep or night-time conditions where cameras fail. Crucially, **sonar does not rely on light** [44†L108-L114] – it works in darkness or murk.
- **Sonar vs. underwater cameras:** Optical cameras are limited by visibility (turbidity, low light) and short range. In contrast, sonar “illuminates” the seafloor by sound and sees through sediment plumes and darkness [44†L108-L114] . AUVs often carry downward-looking video, but these only work in clear, shallow water. Side-scan sonar’s long-range, high-frequency pings can image large areas of bottom that cameras can’t reach. In practice, nearly all deep-sea or large-area debris surveys rely on sonar, not video.
- **Manual inspection problems:** Reviewing thousands of kilometers of sonar logs by eye is **extremely slow and error-prone**. Studies note that existing removal efforts “rely on expert operators and high-cost sonar, limiting coverage” [53†L50-L57] . Operators must scan each image frame for subtle shadows – a tedious “needle in a haystack” task. Fatigue and human error lead to missed nets or false alarms. GhostNetZero notes that net search “has been laborious, usually requiring human review of sonar imagery” [3†L23-L27] . In short, manual sonar analysis doesn’t scale.
- **SIH26057’s goal:** The problem statement asks for an **automated end-to-end pipeline**. In practice, this means building a computer-vision system that ingests raw side-scan sonar images (and their GPS metadata), detects man-made objects on the seafloor, filters out natural clutter, geotags each detection, and presents the results in a user-friendly dashboard [User Prompt] . Core tasks include object detection/segmentation of debris (ghost nets, wrecks, pipes, etc.), a confidence/noise-filter module to reduce false positives, a reporting engine to output latitude/longitude and size of each anomaly, and a UI for uploading imagery and downloading reports.

Essentially, SIH wants a prototype “AI sonar analyst” that can sweep through sonar data much faster than a human.

2. Existing Solutions

- **GhostNetZero (WWF/Microsoft AI for Good):** A research platform aimed at detecting ghost nets via AI [3†L29-L37] . It uses high-resolution shipborne side-scan sonar data and a deep segmentation CNN (DeepLabv3+ResNet50) to find nets. They report ~90% net detection accuracy in test data [3†L29-L37] . The system is human-in-the-loop: experts label initial data, the model learns, and predictions are verified by divers. It runs on Azure GPUs [13†L30-L34] and outputs geo-tagged nets. **Pros:** High accuracy, includes geolocation (every sonar image has lat/lon), active WWF partner network for removals [13†L83-L86] . **Cons:** Not a commercial product, requires specialized hardware/GPU, trained only on certain regions, focuses on nets only.
- **GhostVision (Ping Ecosystem):** An open-source framework (JMSE 2026) for detecting derelict crab pots using low-cost consumer sonar [24†L109-L118] [53†L58-L66] . It includes a public dataset (6,674 images of crab pots in Delaware) [24†L137-L145] . GhostVision trained YOLOv5-style detectors and even a DETR transformer. In field trials, YOLO models achieved recall ~92% and, after tuning, F1 ≈ 0.71 – 0.73 [53†L58-L66] . The pipeline outputs positions of detected “ghost pots” on a map. **Pros:** Open-source; works with cheap fish-finder sonar; demonstrated large-area surveys (1500+ acres) [53†L55-L64] ; relatively fast (10× real-time processing) [53†L61-L64] . **Cons:** Targets only crab pots (a subset of gear), bounding-box only (not masks), recall vs precision trade-offs (YOLO catches most pots but with false alarms [56†L42-L49]), and it lacks a polished UI/dashboard.
- **SonarWiz (Chesapeake Technology):** A **commercial** sidescan post-processing suite. SonarWiz can import many sonar formats (XTF, JSF, etc.), mosaic lines, and capture “contacts” (targets) on sonar logs [15†L22-L30] . It allows manual target editing, PDF/HTML export, and various filters [15†L22-L30] [15†L54-L62] . **Pros:** Industry-standard for hydrographic surveys; supports multi-format sonar; real-time acquisition with quality checks [15†L33-L42] [15†L54-L62] . **Cons:** No built-in AI; target detection is manual or rule-based, not ML. Requires a Windows PC/laptop (not edge). Licensing cost is moderate-high (commercial). It can produce contact reports, but the user must label contacts manually.
- **Klein SonarPro:** A modular sonar acquisition/processing package bundled with Klein side-scan systems [17†L39-L47] . It shows real-time sonar displays, overlays on charts, and provides tools for design and target management [17†L69-L78] . Targets can be measured and logged on the fly. **Pros:** Very user-friendly for sonar operators; integrated with navigation; included free with Klein hardware. **Cons:** Like SonarWiz, no automated AI detection; targets are annotated by human; Windows-only; limited to Klein systems.
- **HYPACK (Xylem software):** A broad hydrographic suite. HYPACK offers side-scan collection, mosaicking, and *targeting* as standard [21†L573-L582] . It can capture “target books” with geo-tagged images and measurements [21†L584-L592] . The “Geocoder” plug-in even does bottom classification. **Pros:** Widely used in surveying, runs full processing (soundings, bathymetry, etc.), and can output formatted reports. **Cons:** No AI-based image analysis: targets must be clicked. The target book is static RTF, not an interactive map or JSON. Licensed product (perpetual or subscription).

- **Other projects:** A few AI research efforts exist. For example, “Ultra-Fast Object Detection in SSS” papers show custom CNNs for sonar objects [54†L6-L8] . Many are academic and focus on one target type (wrecks, mines, etc.). No major company currently markets an AI sonar-cleanup tool specifically for ghost gear.

Summary: Commercial sonar software excels at data management and mosaics but lacks automated debris detection. Research projects like GhostNetZero and GhostVision demonstrate that CNNs/transformers can find nets/pots in sonar imagery [3†L29-L37] [53†L58-L66] . However, existing solutions are either narrow in scope or not production-ready. This project has room to offer a more complete, user-friendly system.

3. Datasets — *Highest Priority*

We located the most relevant public sonar datasets for debris detection:

- **GhostVision (Derelict Crab Pots)** [24†L109-L118] – 6,674 side-scan images collected in Delaware’s bays (consumer Humminbird sonar). Classes: “Crab-Pot” and “Maybe-Crab-Pot” [24†L149-L154] . Annotations: axis-aligned bounding boxes in JSONL (no pixel masks) [24†L146-L154] . No GPS coordinates (annotation is in image pixels). License: CC BY-SA 4.0 [24†L78-L81] (permitting reuse with attribution/share-alike). **Usability:** Excellent for training detection of ghost crab pots; realistic turbidity. Cannot directly teach net-detection, but relevant for net-like structures. **Limitations:** Only crab pots (no true nets, no shipwrecks); shallow, flat environment (bays); image resolutions and sonar frequency fixed; “Maybe” class introduces labeling ambiguity [24†L149-L154] .
- **AI4Shipwrecks (UMich)** [29†L69-L72] – 286 high-resolution SSS images of shipwreck sites (Thunder Bay, Lake Huron, freshwater) with pixelwise masks of wrecks. Sonar: AUV-mounted EdgeTech for lake surveys. Classes: *Shipwreck vs background*. Annotation: binary segmentation masks (pixels) for ship structures [29†L69-L72] . GPS/metadata: likely geo-tagged by AUV but data release focus is on images and masks. Format: PNG images + mask files (see UMich dataset notes). License: CC BY (JMSE dataset) or similar (posted on open research archive). **Usability:** Good for teaching CNNs to spot large anomalies (wrecks), and segmentation. **Limitations:** Very small (286 images); only one object type (wrecks), clean freshwater environment. Masks provided, but no bounding boxes. Not much diversity (all wrecks, mostly large).
- **SeabedObjects-KLSG (Kaggle)** [43†L13-L17] – ~1,190 SSS images compiled for a Kaggle detection challenge. Contains 5 classes: *Shipwreck (385 imgs)*, *Drowning person (36)*, *Plane (62)*, *Mine (129)*, *Seafloor (578)* [43†L13-L17] . Annotations: bounding boxes for each object. License: Kaggle competition (not openly reusable, likely proprietary). **Usability:** Diverse man-made targets vs natural seabed. Good for practicing general sonar object detection. **Limitations:** License restrictions for reuse; classes may not directly include “nets” (though seafloor class is abundant). No nets/pipes.
- **SCTD (Sonar Common Target Dataset)** – Created by Chinese researchers (Ning et al., 2025). Contains various sonar images of generic targets (mines, debris, etc.). Format: Pascal VOC XML annotations. Public on GitHub [51†L208-L216] . **Usability:** Could provide examples of sonar echoes of objects. **Limitations:** No documentation of content count; likely focused on small objects (mines/UXO); not specific to marine debris.

- **SeafloorAI (NeurIPS 2024)** [38†L169-L172] – ~696,000 side-scan images from 62 US surveys (NOAA/USGS). Annotations: pixelwise geological classes (sediments, habitat, faults, etc.) [38†L168-L172] . License: Public domain (NOAA data). **Usability:** Not directly debris, but largest open sonar dataset. Can train models on “natural seafloor” textures to help background understanding or augment data. **Limitations:** No object labels; different classes (rocks, sand, etc.) – useful for pretraining but not directly solving debris detection.
- **CameraSonar and Mine datasets:** Small collections (e.g. 1334 fish images [45†L13-L16] , or specialized mine images [42†L183-L191]). **Use:** Limited; focused on fish or mines in *forward-looking* or specific contexts. Not directly useful for side-scan debris.

Dataset (Year)	# Images	Sonar Type	Classes	BBoxes	Masks	Geo-tags	Format/Ann	License
GhostVision (2026) [24†L137-L145]	6,674	Humminbird SSS	<i>Crab-Pot, Maybe-Crab-Pot</i>	Yes	No	No	JPEG + JSONL (bbox)	CC BY-SA 4.0 [24†L78-L81]
AI4Shipwrecks (2025) [29†L69-L72]	286	EdgeTech SSS (AUV)	<i>Shipwreck</i> (binary vs seafloor)	No	Yes	Likely (via AUV)	PNG images + mask (open on UMich)	CC BY (JMSE)
SeabedObjects KLSG (2024) [43†L13-L17]	1,190	Unknown (survey)	Ship, Plane, Mine, Human, Seafloor	Yes	No	No	(Kaggle CSVs)	Kaggle (proprietary)
SCTD (2025) [51†L208-L216]	~*Unknown	Various SAS	Multiple sonar targets (mines, etc.)	Yes	No	No	Pascal VOC XML	Open (by authors)

Dataset (Year)	# Images	Sonar Type	Classes	BBoxes	Masks	Geo-tags	Format/Ann	License
SeafloorAI (2024) [38†L169-L172]	696,000	NOAA/USGS SSS	Geology (sediment, zones, faults, etc.)	N/A	Yes	Yes	GeoTIFF + masks	Public domain

(“Geo-tags” means each image has associated lat/lon. SCTD geotag uncertain. “Mixable” refers to whether datasets are compatible for joint training – e.g. GhostVision+AI4Shipwrecks both acoustic but different contexts.)

Data gaps: No public datasets include *ghost nets or ropes*, aside from the crab-pot proxy. Also missing: pipe/cylinder, general litter (plastic debris) imagery. We will likely need to collect our own sonar clips or label additional classes. Combining the above: GhostVision (crab pots) + AI4Shipwrecks (wrecks) + maybe SeafloorAI (background) could form a base, but **nets and many debris types remain uncovered**.

4. AI/ML Approaches

We evaluated candidate architectures for sonar target detection:

- **YOLO (You Only Look Once):** A family of one-stage detectors (v5/v7/v8). *Pros:* Very fast (suited to edge/GPU); high recall with tuning [53†L58-L66] . GhostVision found YOLOv5-style models gave high recall (~0.92) on crab pots [53†L59-L66] . YOLO can be trained to produce bounding boxes or even segmentation masks (YOLOv8 supports segmentation outputs). *Cons:* Tends to produce more false positives on sonar (YOLO models caught more true pots but with many FPs [56†L42-L49]). YOLO also needs many annotated examples per class for good performance.
- **Faster R-CNN (and Mask R-CNN):** Two-stage detectors; Mask R-CNN adds pixel masks. *Pros:* Generally higher accuracy for small/rare objects; handles multi-scale better. CNN backbone can be pretrained. Mask R-CNN would directly give segmentation (could be useful for irregular nets). *Cons:* Slower and heavier (likely hard to run on-board); requires careful tuning; training on limited sonar data may overfit. No studies of Mask R-CNN on sonar found in our sources.
- **U-Net / DeepLab (Semantic Segmentation):** CNNs for pixel-wise segmentation. GhostNetZero used DeepLabV3+ with ResNet50 backbone to segment ghost nets [13†L30-L34] . *Pros:* Produces detailed masks, inherently separates background vs anomalies. GhostNetZero reported 94% net detection accuracy using DeepLab [13†L30-L34] . *Cons:* Needs pixel-level labels (costly to annotate); heavy compute, not real-time on edge; only does class vs non-class (no multi-object distinction without extra logic). Also, nets often only occupy few pixels, making segmentation challenging.

- **Transformer-based (DETR, etc.):** Models like DETR treat detection as set prediction. GhostVision's "RF-DETR" achieved the highest precision (fewest false positives) [56†L42-L49] , and had better balanced F1 after filtering. *Pros:* Can perform end-to-end detection, sometimes better for small objects, fewer heuristics. *Cons:* Usually slower to train; requires large datasets. In GhostVision, DETR converged slower but achieved high recall [53†L88-L94] . On edge, DETR is heavy.
- **Others / Sonar-specific:** Some research uses modified CNNs or cascaded detectors for sonar's speckle. For example, local geometric filters to enhance shadow/hilight patterns [54†L6-L8] . But most sonar AI work simply adapts standard vision models with sonar training data. Data scarcity is a key issue: "underwater domains... limited availability of labeled datasets" [32†L123-L131] , so transfer learning is common. Techniques like synthetic data (sonar simulation) can help generate training examples, but was not heavily cited.

Comparison (accuracy vs speed vs data needs): YOLO-style models require bounding-boxes (we have some) and can be fast enough for real-time inference (useful if deploying on an AUV). Faster R-CNN/ Mask-RCNN may get higher precision but likely too slow for edge. U-Net/DeepLab segmentation yields pixel masks and high accuracy (GhostNetZero 94% nets [13†L30-L34]) but demands more GPU. If we only have bounding-box labels (e.g. from GhostVision), object detectors are simpler to train. Transformer models like DETR can reduce false positives but they need *much* data and compute; GhostVision needed frame-to-frame filtering to make it practical [56†L42-L49] .

For **false positives:** Shadow and rocky outcrops cause most FPs. GhostVision showed RF-DETR had far fewer FPs than YOLO [56†L42-L49] . In practice, we'll likely need a secondary filtering stage (e.g. temporal persistence, morphological filters) to cull spurious detections. Confidence-scoring can be learned by a separate classifier or by thresholding.

For **edge deployment:** Lightweight models (YOLO-small, pruned CNNs, quantized networks) are preferred. For example, a YOLOv5-nano or MobileNet backbone U-Net could run on NVIDIA Jetson or Coral TPU. Inference can be batched (per ping line). Training can happen offline on GPU.

Conclusion: There is no one-size-fits-all best model. YOLO or SSD-style detectors are strong candidates for initial prototypes (bounding-box outputs, real-time capable). U-Net/DeepLab segmentation can be explored if we gather mask annotations. Transformer models (DETR) are promising but may be overkill for a student team; they do demonstrate better precision at the cost of complexity [53†L88-L94] [56†L42-L49] . Ultimately, model choice will balance accuracy vs speed vs available training data.

5. Gaps & Opportunity

- **Multi-class detection:** GhostVision and GhostNetZero focus on one object class (pots or nets). No existing tool detects *multiple* debris types (nets, ropes, cylinders, pipes, wrecks) simultaneously. Building a multi-class model (e.g. nets + nets + wrecks + trash) is an opportunity. Even labeling some common background objects (rocks, natural shadows) as negatives could reduce mistakes.
- **False-positive reduction:** As GhostVision's study shows, even state-of-the-art models make many FPs on sonar [56†L42-L49] . There is a gap in robust filtering: e.g. temporal consistency (requiring detections across multiple ping strips), or physics-based filters. SIH26057 mentions a

“noise filtering” module. A novel approach (e.g. combining acoustic echo profiles or a second CNN to reject false candidates) could set our solution apart.

- **Confidence and risk scoring:** No existing system provides a *risk/confidence score* beyond raw probability. We could design a scoring scheme (0–100%) that accounts for sonar quality, target ambiguity, etc. For instance, flag low-confidence detections for human review. Or compute a “net entanglement risk” for each detection by analyzing shape features.
- **Geolocation & reporting:** Tools like HYPACK capture target positions, but no standard pipeline outputs a structured JSON/CSV report of detections. GhostNetZero and GhostVision use metadata to geotag nets [13†L83-L86] , but do not produce user-ready report files. SIH specifically wants lat/lon and size in a report. Automating this (parsing the sonar log’s coordinate data) and packaging it with images on a map is a needed feature.
- **User verification and UI:** Existing AI pipelines either leave humans out (pure automation) or rely entirely on experts. There’s room for a “human-in-the-loop” dashboard that shows detected objects on a map and lets users approve/reject. An easy-to-use front-end (desktop or web app) for uploading raw data and visualizing results in real time is largely missing in current products. Even a basic GUI that overlays detections on a mosaic (see [15]) would be valuable for end users (surveyors, conservationists).
- **Edge/onboard deployment:** GhostNetZero is cloud-based; GhostVision demos ran on laptops. No solution is packaged for onboard use on an AUV or drone. Optimizing the pipeline to run on a Jetson Nano or similar (e.g. TensorRT-accelerated YOLO) could be a differentiator.

3–5 Realistic differentiators for our SIH team:

1. **Multi-Object Detection:** Train a unified model for *multiple debris classes* (nets, pots, wrecks, pipes, etc.) rather than one-at-a-time. This addresses a gap in both GhostNetZero and GhostVision.
2. **Active False-Positive Filtering:** Implement post-processing such as *temporal persistence* (detections only count if seen in >N consecutive pings) or ensemble voting to slash false alarms [56†L42-L49] . Such a filter could dramatically improve real-world usability.
3. **Confidence-Scored Alerts:** Combine model confidence with sonar image quality metrics (e.g. signal-to-noise, shadow length) into a single *risk score* for each detection. Users get not just “found an object” but “80% sure this is a net”.
4. **Lightweight Edge-Ready Pipeline:** Design the whole stack (detection + filtering + geotagging) to run on a small GPU (e.g. Jetson) with offline mode. This would enable on-vehicle real-time use, unlike cloud-bound solutions.
5. **Integrated Dashboard with Reporting:** Even a simple web/mobile interface that ties it all together (upload data → process → show map of anomalies → export CSV/JSON) would be innovative. Existing solutions lack an end-to-end user interface targeted at non-experts.

6. Business and India Context

- **Potential customers:**

- *Government agencies:* Coast Guard, Navy, Ports (Mumbai, Chennai, Kochi etc.) and Port Trusts could use sonar debris detection to keep harbors safe. Ministry of Earth Sciences (MoES) or Marine Fisheries Dept may fund surveys. Environmental bodies (e.g. MoEFCC) and Indian Council of Agricultural Research (ICAR) institutes like CMFRI, CIFT monitor marine pollution.
- *Survey companies:* Indian hydrographic firms (e.g. Survey India, NIOT contractors) and offshore oil & gas (ONGC, Reliance) doing seabed surveys might purchase analysis software. Offshore windfarm developers (growing in India) also map seabeds.
- *Fishing/fisheries:* Port communities and large fishing co-ops concerned about nets might use a service to map ghost gear in their areas. NGOs (HCL Foundation, WWF-India, local blue economy groups) running cleanup projects may use such tools.
- *Commercial salvage & cleanup:* Companies clearing harbors or wrecks could use automated sonar analysis to plan missions.

- *Academia/research:* Institutions like NIOT, NIO, IIT Madras etc. might use an analysis tool for marine research data.

- **Use cases:** E.g. periodic surveying of fishing harbors to locate lost nets (helping fishermen recover gear and reduce ghost fishing), or offshore pipeline inspections to spot entangled nets. Search-and-rescue teams might detect bodies or debris fields (if such classes are included).

- **Revenue models:**

- *SaaS platform:* Host a cloud/edge service where clients upload SSS logs and get back analysis (pay per hour of sonar, or monthly subscription).
- *One-time software license:* Sell the software suite (with per-seat or per-vessel licensing) to government agencies or companies.
- *Consulting/Contracting:* Offer end-to-end survey+analysis packages (for example, priced per survey mile) – governments and ports often contract out hydro surveys.
- *Government grants/CSR:* Many ghost gear projects are funded by environment programs, so partial funding by government R&D or NGO grants is possible.
- *Hardware bundle:* Partner with sonar manufacturers to embed our analytics (like a value-add to Edgetech/Kongsberg sonars).

- **India-specific insights:** India's 11,098 km coastline faces a **growing ghost gear crisis** [63†L79-L87] . Recent media (Aug 2026) highlight that lost nets account for ~10% of marine litter globally [63†L84-L92] and Indian states (e.g. Tamil Nadu) are setting up collection centers [63†L161-L170] . However, India lacks a national ALDFG (abandoned gear) framework [63†L75-L83] . This means early pilot opportunities: e.g. collaborate with Tamil Nadu Pollution Control Board, Gujarat Fisheries Dept, or regional sea turtle conservation NGOs, to run feasibility studies.

Government initiatives like Pradhan Mantri Matsya Sampada Yojana (PMMSY) emphasize sustainable fisheries, so technology for gear retrieval aligns with policy. The Indian Navy/Coast Guard might also be interested in removing navigation hazards. On the dataset side, Indian institutions may have unpublished sonar logs (e.g. ICAR-NIO surveys) – forming pilot data partnerships could benefit both sides.

Summary Decision Points (A-J):

- **A. What SIH wants:** An AI-driven system that automatically processes side-scan sonar logs to find and report man-made seabed debris (ghost nets, wrecks, pipes, etc.). It should output geo-tagged locations and basic classification of each anomaly, presented through a simple UI. Key parts: object detection/segmentation, false-positive filtering with confidence scores, geolocation parsing, and a user dashboard for uploads and report downloads.
- **B. What already exists:** Specialized solutions exist for *single* tasks. GhostNetZero and GhostVision demonstrate that CNNs can detect nets/pots in sonar [3†L29-L37] [53†L58-L66] . Commercial sonar suites (HYPACK, SonarWiz, SonarPro) provide data handling and contact marking [15†L22-L30] [17†L69-L78] , but no automated ML detection. No current tool does the full SIH pipeline.
- **C. Best datasets available:** Public training data is scarce. The **GhostVision** crab-pot dataset (6,674 images) [24†L137-L145] and **AI4Shipwrecks** (286 images with wreck masks) [29†L69-L72] are the most relevant open resources. The SeabedObjects-KLSG challenge data [43†L13-L17] can augment detection skills (mines, wrecks, people). Large background datasets like **SeafloorAI** (696k images of seabed geology) [38†L169-L172] are useful for pretraining or noise modeling.
- **D. What data we are missing:** **Ghost nets and general marine debris** images are lacking. Current sets cover crab-pots, wrecks, mines, but *no public side-scan data of abandoned nets, cylinders, pipes, or synthetic litter*. Also missing are paired geolocation-tagged sonar logs. We may need to capture our own data (e.g. partner with fisheries dept or use consumer sonar) to train for true net shapes.
- **E. Hardest technical challenges:** (1) Sonar imagery is very noisy: speckle noise and uneven illumination make small/flat nets hard to distinguish from rocks. (2) Data scarcity: only a few thousand annotated images total, far below typical computer vision datasets [32†L123-L131] . (3) High false positives: natural features (shadow bands, boulders) often look like debris, so reducing FPs is tricky [56†L42-L49] . (4) Varying conditions: different sonars, frequencies, and survey setups (boat vs AUV) can change appearances. (5) On-device efficiency: running deep models on an underwater drone's limited hardware.
- **F. Student team scope:** A feasible project is to build a prototype that **demonstrates automated detection** on a few classes. For example, training YOLO on the available GhostVision and Shipwreck datasets to find crab-pots and wrecks, with a post-filter to weed out obvious false detections. We can integrate simple GPS parsing and output (e.g. GeoJSON). A web or Jupyter dashboard can allow uploading sample sonar images and displaying results. We should **not** promise large-scale deployment or a full multi-platform app yet, but rather a proof-of-concept pipeline with basic UI.
- **G. Strongest differentiators:** Based on the gaps, our solution could stand out by (1) **handling multiple debris types** simultaneously, (2) incorporating a **robust false-positive filter** (e.g. leveraging temporal consistency), (3) providing **quantitative confidence scores** and (4) targeting **low-cost hardware** for edge use. A user-friendly interface (with options like Hindi language support) and ready-to-use report generation (CSV/JSON) would also be novel for this domain.

- **H. Best potential customers:** In India: the *Fisheries Departments* of coastal states, *ICAR marine institutes* (CMFRI, CIFT) and *NIO* for monitoring; *Port authorities* (Chennai, Mumbai, Kochi) for harbor maintenance; *ONGC/Offshore wind companies* for offshore surveys; *Naval operations* (harbor security). Globally: hydrographic survey firms, marine conservancies (WWF, Ocean Cleanup), and energy companies.
- **I. Realistic business models:** Likely a mix of **government contracts and SaaS**. Initially, partnering on government-sponsored pilot surveys (fishing harbors, turtle nesting areas) could demonstrate value. Eventually, a subscription platform (per-survey or per-area pricing) or perpetual licenses for agencies. CSR/NGO grants (as seen with WWF's GhostNetZero) might help fund early development.
- **J. Next steps:** Start by *collecting/curating data*. Acquire any available side-scan samples (even synthetic or from hobbyist sonar). Experiment with training a detector on GhostVision/Shipwreck data to validate feasibility. Prototype the pipeline stages (preprocessing, detection, geotagging, report generation) in parallel. Involve end-users early (e.g. NIOT or fishermen groups) for feedback on reports/UI. Finally, iterate on the model (reducing FPs) and begin integration into a simple demo interface. These steps will lay groundwork for a competitive prototype meeting SIH26057's goals.

Sources: We synthesized information from academic and industry sources. Key references include WWF/Microsoft's GhostNetZero work [3†L23-L27] [13†L30-L34] , the GhostVision JMSE paper [53†L58-L66] , NOAA definitions [9†L12-L17] , and sonar software documentation [15†L22-L30] [21†L573-L582] , among others. These grounded our understanding of ghost gear issues, existing tech, datasets, and model performance.
